

LAPORAN
WORKSHOP SISTEM OPERASI



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Package Linux

A. PERCOBAAN

a. Percobaan 1:

Sebelum instalasi modul, pastikan sistem diperbarui:

\$ su -

\$ sudo apt update

\$ sudo apt upgrade -y

```
wirda@Wirda2SICUbuntu:~/Desktop$ cd
wirda@Wirda2SICUbuntu:~$ su -
Password:
root@Wirda2SICUbuntu:~# sudo apt update
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:2 http://id.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://id.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://id.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 http://id.archive.ubuntu.com/ubuntu jammy-updates/main i386 Packages [912 kB]
Get:6 http://id.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,065 kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2,818 kB]
Get:8 http://id.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [473 kB]
Get:9 http://id.archive.ubuntu.com/ubuntu jammy-updates/main amd64 DEP-11 Metadata [112 kB]
Get:10 http://id.archive.ubuntu.com/ubuntu jammy-updates/main amd64 C-N-F Metadata [19.0 kB]
Get:11 http://id.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [4,832 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [732 kB]
```

```
root@Wirda2SICUbuntu:~# sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
  gssapi-common libgssapi7
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following NEW packages will be installed:
  linux-headers-6.8.0-87-generic linux-hwe-6.8-headers-6.8.0-87
  linux-hwe-6.8-tools-6.8.0-87 linux-image-6.8.0-87-generic
  linux-modules-6.8.0-87-generic linux-modules-extra-6.8.0-87-generic
  linux-tools-6.8.0-87-generic ubuntu-pro-client ubuntu-pro-client-l10n
The following packages have been kept back:
  libnss-systemd libpam-systemd libsystemd0 libudev1 systemd systemd-oom
  systemd-sysv systemd-timesyncd udev
The following packages will be upgraded:
  alsa-ucm-conf apt apt-utils base-files bind9-dnsutils bind9-host bind9-libs
  coreutils dconf-cli dconf-gsettings-backend dconf-service distro-info
  distro-info-data dmidecode dmsetup dns-root-data evince evince-common
  firmware-sof-signed fonts-noto-color-emoji gdm3 gir1.2-gdm-1.0
  gir1.2-mutter-10 gir1.2-nm-1.0 gir1.2-packagekit-glib-1.0 gjs
```

- **\$ su -** : Perintah ini digunakan untuk masuk ke mode superuser (root). Setelah dijalankan, sistem meminta password. Jika berhasil, prompt akan berubah menjadi root@....
- **\$ sudo apt update** : Perintah ini berfungsi untuk memperbarui daftar paket dari repository Ubuntu. Dari output terlihat bahwa sistem mengambil data dari beberapa sumber repository seperti: security.ubuntu.com, id.archive.ubuntu.com. Proses ini hanya memperbarui informasi paket, bukan meng-upgrade pakatnya.

- **\$ sudo apt upgrade -y** : Perintah ini digunakan untuk meng-upgrade seluruh paket yang mempunyai versi terbaru. Opsi -y digunakan agar sistem otomatis menjawab yes pada semua konfirmasi.

b. Percobaan 2:

Instalasi Paket Perangkat Lunak

\$ sudo apt install apache2 -y

```
root@Wirda2SICubuntu:~# sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-79-generic linux-hwe-6.8-headers-6.8.0-79
  linux-hwe-6.8-tools-6.8.0-79 linux-image-6.8.0-79-generic
  linux-modules-6.8.0-79-generic linux-modules-extra-6.8.0-79-generic
  linux-tools-6.8.0-79-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap
0 upgraded, 8 newly installed, 0 to remove and 9 not upgraded.
Need to get 1,923 kB of archives.
After this operation, 7,728 kB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64
1.7.0-8ubuntu0.22.04.2 [108 kB]
Get:2 http://id.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 a
```

- **\$ sudo apt install apache2 -y** : Perintah untuk menginstal web server Apache2 secara otomatis tanpa meminta konfirmasi pengguna.

c. Percobaan 3:

Instalasi Modul PHP:

\$ sudo apt install php libapache2-mod-php php-mysql php-xml php-curl php-mbstring -y

```
root@Wirda2SICubuntu:~# sudo apt install php libapache2-mod-php php-mysql php-xml
l php-curl php-mbstring -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-79-generic linux-hwe-6.8-headers-6.8.0-79
  linux-hwe-6.8-tools-6.8.0-79 linux-image-6.8.0-79-generic
  linux-modules-6.8.0-79-generic linux-modules-extra-6.8.0-79-generic
  linux-tools-6.8.0-79-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libapache2-mod-php8.1 libonig5 php-common php8.1 php8.1-cli php8.1-common
  php8.1-curl php8.1-mbstring php8.1-mysql php8.1-opcache php8.1-readline
  php8.1-xml
Suggested packages:
  php-pear
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.1 libonig5 php php-common php-curl
  php-mbstring php-mysql php-xml php8.1 php8.1-cli php8.1-common php8.1-curl
  php8.1-mbstring php8.1-mysql php8.1-opcache php8.1-readline php8.1-xml
0 upgraded, 18 newly installed, 0 to remove and 9 not upgraded.
```

- **\$ sudo apt install php libapache2-mod-php php-mysql php-xml php-curl php-mbstring -y** : Menginstal PHP beserta modul-modul

pendukungnya seperti modul Apache, MySQL, XML, CURL, dan MBString untuk menjalankan aplikasi web berbasis PHP.

d. Percobaan 4:

Instalasi Modul Database (MySQL):

\$ sudo apt install mysql-server -y

```
root@Wirda2SICUbuntu:~# sudo apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-79-generic linux-hwe-6.8-headers-6.8.0-79
  linux-hwe-6.8-tools-6.8.0-79 linux-image-6.8.0-79-generic
  linux-modules-6.8.0-79-generic linux-modules-extra-6.8.0-79-generic
  linux-tools-6.8.0-79-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libaio1 libbcgi-fast-perl libbcgi-pm-perl libevent-core-2.1-7
  libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl libfcgi0ldbl
  libhtml-template-perl libmecab2 libprotobuf-lite23 mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0
  mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libipc-sharedcache-perl tinycb
The following NEW packages will be installed:
  libaio1 libbcgi-fast-perl libbcgi-pm-perl libevent-core-2.1-7
  libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl libfcgi0ldbl
  libhtml-template-perl libmecab2 libprotobuf-lite23 mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0
```

- \$ sudo apt install mysql-server -y : Menginstal MySQL Server secara otomatis tanpa meminta konfirmasi pengguna.

e. Percobaan 5:

Modul kernel digunakan untuk menambahkan dukungan terhadap perangkat keras atau fitur tertentu tanpa harus mengubah kernel utama:

Gunakan modprobe untuk memuat modul kernel.

\$ sudo modprobe ext4

Gunakan perintah berikut untuk melihat modul kernel yang sedang digunakan:

\$ lsmod

```
root@Wirda2SICUbuntu:~# sudo apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-79-generic linux-hwe-6.8-headers-6.8.0-79
  linux-hwe-6.8-tools-6.8.0-79 linux-image-6.8.0-79-generic
  linux-modules-6.8.0-79-generic linux-modules-extra-6.8.0-79-generic
  linux-tools-6.8.0-79-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libaio1 libbcgi-fast-perl libbcgi-pm-perl libevent-core-2.1-7
  libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl libfcgi0ldbl
  libhtml-template-perl libmecab2 libprotobuf-lite23 mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0
  mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libipc-sharedcache-perl tinycb
The following NEW packages will be installed:
  libaio1 libbcgi-fast-perl libbcgi-pm-perl libevent-core-2.1-7
  libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl libfcgi0ldbl
  libhtml-template-perl libmecab2 libprotobuf-lite23 mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0
```

- **\$ sudo modprobe ext4** : Memuat modul kernel ext4 secara manual untuk menambah dukungan filesystem tanpa perlu restart kernel.
- **\$ lsmod** : Menampilkan daftar modul kernel yang sedang aktif/terload di sistem.

f. Percobaan 6:

Verifikasi dan Konfigurasi Modul

Memeriksa Instalasi Paket, Gunakan dpkg untuk memastikan modul sudah terpasang:

\$ dpkg -l | grep apache2

```
root@Wirda2SICUbuntu:~# dpkg -l | grep apache2
ii apache2 2.4.52-1ubuntu4.16
    amd64 Apache HTTP Server
ii apache2-bin 2.4.52-1ubuntu4.16
    amd64 Apache HTTP Server (modules and other binary files)
ii apache2-data 2.4.52-1ubuntu4.16
    all Apache HTTP Server (common files)
ii apache2-utils 2.4.52-1ubuntu4.16
    amd64 Apache HTTP Server (utility programs for web servers)
ii libapache2-mod-php 2:8.1+92ubuntu1
    all server-side, HTML-embedded scripting language (Apache 2 modu
le) (default)
ii libapache2-mod-php8.1 8.1.2-1ubuntu2.22
    amd64 server-side, HTML-embedded scripting language (Apache 2 modu
le)
```

- **\$ dpkg -l | grep apache2** : Perintah ini digunakan untuk memastikan bahwa paket apache2 sudah terpasang di dalam sistem. Sesuai output paket apache2 telah terinstal, seperti: apache2, apache2-bin, apache2-data, apache2-utils, dan modul pendukung lainnya.

g. Percobaan 7:

Gunakan systemctl untuk memeriksa apakah layanan berjalan dengan baik:

\$ sudo systemctl status apache2

\$ sudo systemctl status mysql

```
root@Wirda2SICUbuntu:~# sudo systemctl status apache2
* apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor prese
   Active: active (running) since Tue 2025-11-18 09:58:15 +07; 10min ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 45409 (apache2)
      Tasks: 6 (limit: 2264)
    Memory: 14.2M
       CPU: 150ms
    CGroup: /system.slice/apache2.service
            |-45409 /usr/sbin/apache2 -k start
            |-45411 /usr/sbin/apache2 -k start
            |-45412 /usr/sbin/apache2 -k start
            |-45413 /usr/sbin/apache2 -k start
            |-45414 /usr/sbin/apache2 -k start
            `--45415 /usr/sbin/apache2 -k start

<<9E>.U. 18 09:58:15 Wirda2SICUbuntu systemd[1]: Starting The Apache HTTP Se
<<9E>.U. 18 09:58:15 Wirda2SICUbuntu systemd[1]: Started The Apache HTTP Ser
lines 1-18/18 (END)
[1]+  Stopped                  sudo systemctl status apache2
```

```
root@Wirda2SICUbuntu:~# sudo systemctl status mysql
* mysql.service - MySQL Community Server
   Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset:
   Active: active (running) since Tue 2025-11-18 09:59:59 +07; 9min ago
   Process: 46046 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=
   Main PID: 46054 (mysqld)
     Status: "Server is operational"
    Tasks: 37 (limit: 2264)
   Memory: 364.5M
      CPU: 6.740s
   CGroup: /system.slice/mysql.service
           └─46054 /usr/sbin/mysqld

<<9E>.B. 18 09:59:58 Wirda2SICUbuntu systemd[1]: Starting MySQL Community Se
<<9E>.B. 18 09:59:59 Wirda2SICUbuntu systemd[1]: Started MySQL Community Ser
lines 1-14/14 (END)
[2]+  Stopped                  sudo systemctl status mysql
```

- **\$ sudo systemctl status apache2** : Perintah tersebut menampilkan status layanan Apache2. Jika layanan berjalan, maka akan muncul status: active (running).
- **\$ sudo systemctl status mysql** : Jika layanan berjalan normal, maka akan muncul informasi: active (running) beserta detail seperti waktu aktif, path service, dan log aktivitas.

B. PERCOBAAN MANAJEMEN ARSIP DAN KOMPRESI

Tar (Tape Archive) digunakan untuk menggabungkan banyak file atau direktori menjadi satu berkas arsip .tar. gzip digunakan untuk mengompresi (mengecilkan ukuran) file, biasanya menghasilkan file dengan ekstensi .gz.

a. Percobaan 1:

Buat direktori kerja khusus untuk praktikum di root:

\$ su -

\$ mkdir arsip

\$ cd ./arsip

```
Wirda@Wirda2SICUbuntu:~$ su -
Password:
root@Wirda2SICUbuntu:~# mkdir arsip
root@Wirda2SICUbuntu:~# cd ./arsip
root@Wirda2SICUbuntu:~/arsip#
```

- **\$ su** : Perintah untuk masuk ke mode superuser (root). Setelah dijalankan, kamu akan diminta memasukkan password root.
- **\$ mkdir arsip** : Membuat direktori baru bernama arsip di lokasi saat ini.
- **\$ cd ./arsip** : Masuk ke direktori arsip yang baru saja dibuat.

b. Percobaan 2:

Gunakan tar untuk membuat arsip konfigurasi:

\$ sudo tar -cvf etc-backup.tar /etc

```

root@Wirda2SICUbuntu:~/arsip# sudo tar -cvf etc-backup.tar /etc
tar: Removing leading '/' from member names
/etc/
/etc/PackageKit/
/etc/PackageKit/PackageKit.conf
/etc/PackageKit/Vendor.conf
/etc/gai.conf
/etc/os-release
/etc/protocols
/etc/cron.daily/
/etc/cron.daily/appport
/etc/cron.daily/apt-compat
/etc/cron.daily/apache2
/etc/cron.daily/dpkg
/etc/cron.daily/logrotate
/etc/cron.daily/anaconda
/etc/cron.daily/man-db
/etc/cron.daily/cracklib-runtime
/etc/cron.daily/.placeholder
/etc/skel/
/etc/skel/.profile
/etc/skel/.bashrc
/etc/skel/.bash_logout

```

- **\$ sudo tar -cvf etc-backup.tar /etc** : Membuat arsip (backup) bernama etc-backup.tar berisi seluruh isi direktori /etc. Opsi c : create (membuat arsip baru).v : verbose (menampilkan proses yang sedang berjalan). f : file (menentukan nama file arsip yang akan dibuat).

c. Percobaan 3:

Kompres hasil arsip agar lebih kecil:

\$ sudo gzip etc-backup.tar

Mengecek ukuran file

\$ ls -lh etc-backup.tar.gz

```

root@Wirda2SICUbuntu:~/arsip# sudo gzip etc-backup.tar

```

```

root@Wirda2SICUbuntu:~/arsip# ls -lh etc-backup.tar.gz
-rw-r--r-- 1 root root 1.5M W. 18 10:17 etc-backup.tar.gz

```

- **\$ sudo gzip etc-backup.tar** : Perintah ini digunakan untuk mengompres file arsip etc-backup.tar menggunakan algoritma gzip. Setelah dikompres, ekstensi file berubah menjadi .tar.gz.
- **\$ ls -lh etc-backup.tar.gz** : Perintah ini digunakan untuk mengecek ukuran file setelah dikompresi. Hasilnya menunjukkan bahwa ukuran file menjadi lebih kecil daripada file .tar aslinya.

d. Percobaan 4:

Membuat Arsip dan Kompres Sekaligus, Gunakan tar langsung dengan opsi z untuk kompresi gzip:

\$ sudo tar -czvf varlog-backup.tar.gz /var/log

Mengecek Isi Arsip Tanpa Mengekstrak:

\$ tar -tzvf varlog-backup.tar.gz | less

Tekan CTRL+Z untuk menghentikan proses

Ketikan ls untuk melihat isi arsip


```

root@Wirda2SICUbuntu:~/arsip# sudo tar -czvf varlog-backup.tar.gz /var/log
tar: Removing leading '/' from member names
/var/log/
/var/log/dmesg.3.gz
/var/log/mysql/
/var/log/mysql/error.log
/var/log/kern.log.1
/var/log/mail.log.2.gz
/var/log/dpkg.log.2.gz
/var/log/auth.log.3.gz
/var/log/dmesg.1.gz
/var/log/auth.log.4.gz
/var/log/syslog.2.gz
/var/log/mail.log
/var/log/speech-dispatcher/
/var/log/boot.log.5
/var/log/ubuntu-advantage.log.2.gz
/var/log/boot.log.6
/var/log/boot.log
/var/log/mail.log.4.gz
/var/log/vboxpostinstall.log
/var/log/gdm3/
/var/log/faillog

```

```

root@Wirda2SICUbuntu:~/arsip# tar -tzvf varlog-backup.tar.gz | less
[3]+  Stopped                  tar -tzvf varlog-backup.tar.gz | less

```

```

drwxrwxr-x root/syslog      0 2025-11-18 09:59 var/log/
-rw-r----- root/adm       14743 2025-10-21 09:53 var/log/dmesg.3.gz
drwxr-x-- mysql/adm        0 2025-11-18 09:59 var/log/mysql/
-rw-r----- mysql/adm     4292 2025-11-18 09:59 var/log/mysql/error.log
-rw-r----- syslog/adm   136870 2025-11-18 09:29 var/log/kern.log.1
-rw-r----- syslog/adm    162 2025-11-04 10:05 var/log/mail.log.2.gz
-rw-r----- root/root   124067 2025-09-30 11:58 var/log/dpkg.log.2.gz
-rw-r----- syslog/adm    1190 2025-11-04 10:05 var/log/auth.log.3.gz
-rw-r----- root/adm    14526 2025-11-11 07:15 var/log/dmesg.1.gz
-rw-r----- syslog/adm    1256 2025-10-21 09:53 var/log/auth.log.4.gz
-rw-r----- syslog/adm   38304 2025-11-11 07:15 var/log/syslog.2.gz
-rw-r----- syslog/adm     617 2025-11-18 09:48 var/log/mail.log
drwx----- speech-dispatcher/root 0 2023-05-23 14:29 var/log/speech-dispatcher/
-rw----- root/root      8578 2025-10-14 09:08 var/log/boot.log.5
-rw-r--r-- root/root      1971 2025-10-07 10:34 var/log/ubuntu-advantage.log.2.gz
-rw----- root/root     10530 2025-10-07 09:56 var/log/boot.log.6
-rw----- root/root        0 2025-11-18 09:29 var/log/boot.log
-rw-r----- syslog/adm     166 2025-10-14 09:16 var/log/mail.log.4.gz
-rw-rw-r-- root/root       890 2025-09-16 11:19 var/log/vboxpostinstall.log
drwx-x-x-x root/gdm        0 2025-09-16 11:21 var/log/gdm3/
-rw-r--r-- root/root     32064 2025-11-18 09:59 var/log/faillog
-rw-r--r-- root/root     42115 2025-10-21 10:01 var/log/dpkg.log.1
:

```

```

root@Wirda2SICUbuntu:~/arsip# ls
etc-backup.tar.gz  varlog-backup.tar.gz

```

- **\$ sudo tar -czvf varlog-backup.tar.gz /var/log** : Perintah ini membuat arsip sekaligus mengompresnya dalam format gzip. Opsi: c = create. z = kompresi gzip. v = verbose (menampilkan proses). f = nama file. Hasilnya adalah file arsip bernama varlog-backup.tar.gz yang berisi seluruh file dalam direktori /var/log.
- **\$ tar -tzvf varlog-backup.tar.gz | less** : Perintah ini digunakan untuk melihat isi arsip tanpa melakukan ekstrak. Dengan tambahan | less, tampilan bisa discroll hingga bawah. Jika proses terlalu panjang, kamu bisa menghentikannya dengan CTRL+Z.
- **\$ ls** : Perintah ini digunakan untuk menampilkan daftar file dalam direktori saat ini, dan akan menampilkan file arsip yang baru dibuat.

e. **Percobaan 5:**

Membuat Arsip

\$ sudo tar -cvf usr-local-backup.tar /usr/local/bin

Kompres dengan gzip

\$ sudo gzip usr-local-backup.tar

```
root@Wirda2SICUbuntu:~/arsip# sudo tar -cvf usr-local-backup.tar /usr/local/bin
tar: Removing leading '/' from member names
/usr/local/bin/
```

```
root@Wirda2SICUbuntu:~/arsip# sudo gzip usr-local-backup.tar
root@Wirda2SICUbuntu:~/arsip# ls
etc-backup.tar.gz  usr-local-backup.tar.gz  varlog-backup.tar.gz
```

- **\$ sudo tar -cvf usr-local-backup.tar /usr/local/bin** : Perintah ini membuat arsip biasa (tanpa kompresi) bernama usr-local-backup.tar yang berisi seluruh file dalam direktori /usr/local/bin.
- **\$ sudo gzip usr-local-backup.tar** : Perintah ini mengompres file arsip menjadi format gzip. Setelah proses selesai, file berubah menjadi usr-local-backup.tar.gz.
- **\$ ls** : Perintah ini digunakan untuk memastikan file arsip dan file terkompresi berhasil dibuat, misalnya: usr-local-backup.tar.gz, varlog-backup.tar.gz

C. PERCOBAAN EKSTRAKSI ARSIP

a. **Percobaan 1:**

Buat direktori tujuan:

\$ mkdir restore-tes

\$ cd restore-tes

```
root@Wirda2SICUbuntu:~/arsip# cd
root@Wirda2SICUbuntu:~# mkdir restore-tes
root@Wirda2SICUbuntu:~# cd ./restore-tes
root@Wirda2SICUbuntu:~/restore-tes# sudo tar -xvzf /root/arsip/etc-backup.tar.gz
etc/
etc/PackageKit/
etc/PackageKit/PackageKit.conf
etc/PackageKit/Variant.conf
etc/gai.conf
etc/os-release
etc/protocols
etc/cron.daily/
etc/cron.daily/apport
etc/cron.daily/apt-compat
etc/cron.daily/apache2
etc/cron.daily/dpkg
etc/cron.daily/logrotate
etc/cron.daily/anaconda
etc/cron.daily/man-db
etc/cron.daily/cracklib-runtime
etc/cron.daily/placeholder
etc/skel/
etc/skel/.profile
etc/skel/.bashrc
```

- **\$ mkdir restore-tes** dan **\$ cd restore-tes** : Perintah ini membuat folder bernama restore-tes, lalu berpindah ke dalamnya untuk melakukan proses ekstraksi backup.

- **\$ sudo tar -xzf /root/arsip/etc-backup.tar.gz** : Perintah ini digunakan untuk mengekstrak file backup bernama etc-backup.tar.gz. Opsi yang digunakan: x = extract (mengeksktrak). z = gzip (karena file .gz). v = verbose (menampilkan proses ekstraksi). f = nama file arsip. Hasilnya: semua file dari backup /etc diekstrak ke dalam direktori restore-tes.

b. Percobaan 2:

Mengekstrak file konfigurasi yang dibackup sebelumnya:

\$ sudo tar -xzf /root/arsip/etc-backup.tar.gz

Verifikasi Hasil Ekstraksi:

\$ ls

\$ ls -l etc/

```
root@Wirda2SICubuntu:~/restore-tes# ls
etc
root@Wirda2SICubuntu:~/restore-tes# ls -l etc/
total 1264
drwxr-xr-x  3 root root   4096 2023  acpi
-rw-r--r--  1 root root   3028 2023  adduser.conf
-rw-r--r--  1 root root    51 2023  09:16  aliases
-rw-r--r--  1 root root  12288 2023  09:16  aliases.db
drwxr-xr-x  3 root root   4096 2023  alsa
drwxr-xr-x  2 root root   4096 2023  09:59  alternatives
-rw-r--r--  1 root root    335 2022  11:23  anacrontab
drwxr-xr-x  8 root root   4096 2023  09:51  apache2
-rw-r--r--  1 root root    433 2022  11:23  app.config
drwxr-xr-x  5 root root   4096 2023  apm
drwxr-xr-x  3 root root   4096 2023  11:14  apparmor
drwxr-xr-x  8 root root   4096 2023  09:59  apparmor.d
drwxr-xr-x  4 root root   4096 2023  12:20  appport
-rw-r--r--  1 root root    769 2022  09:22  appstream.conf
drwxr-xr-x  8 root root   4096 2023  11:18  apt
drwxr-xr-x  3 root root   4096 2023  12:21  avahi
-rw-r--r--  1 root root   2319 2022  11:06  bash.bashrc
-rw-r--r--  1 root root    45 2021  11:11  bash_completion
drwxr-xr-x  2 root root   4096 2023  12:20  bash_completion.d
-rw-r--r--  1 root root    367 2020  11:16  bindresvport.blacklist
```

- **\$ ls** : Perintah ini menampilkan daftar file/direktori yang sudah berhasil diekstrak.
- **\$ ls -l etc/** : Perintah ini menampilkan isi folder etc menggunakan format long listing, sehingga terlihat: permission file, pemilik, ukuran, waktu modifikasi, nama file/direktori, Ini membuktikan bahwa file backup berhasil diekstraksi dengan lengkap.

LATIHAN

1. Lakukan instalasi paket Nginx sebagai alternatif Apache, kemudian periksa apakah layanan Nginx sudah aktif di sistem.

```
root@Wirda2SICUbuntu:~/restore-tes# cd
root@Wirda2SICUbuntu:~# sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-79-generic linux-hwe-6.8-headers-6.8.0-79
  linux-hwe-6.8-tools-6.8.0-79 linux-image-6.8.0-79-generic
  linux-modules-6.8.0-79-generic linux-modules-extra-6.8.0-79-generic
  linux-tools-6.8.0-79-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libnginx-mod-http-geoip2 libnginx-mod-http-image-filter
  libnginx-mod-http-xslt-filter libnginx-mod-mail libnginx-mod-stream
  libnginx-mod-stream-geoip2 nginx-common nginx-core
Suggested packages:
  fcgiwrap nginx-doc
The following NEW packages will be installed:
  libnginx-mod-http-geoip2 libnginx-mod-http-image-filter
  libnginx-mod-http-xslt-filter libnginx-mod-mail libnginx-mod-stream
```

```
root@Wirda2SICUbuntu:~# sudo systemctl status nginx
x nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset:
   Active: failed (Result: exit-code) since Tue 2025-11-18 10:38:27 +07; 33s
   Docs: man:nginx(8)
   Process: 46657 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_pro
   Process: 46658 ExecStart=/usr/sbin/nginx -g daemon on; master_process on;
   CPU: 18ms

<<[9E]> . 18 10:38:25 Wirda2SICUbuntu nginx[46658]: nginx: [emerg] bind() to
<<[9E]> . 18 10:38:25 Wirda2SICUbuntu nginx[46658]: nginx: [emerg] bind() to
<<[9E]> . 18 10:38:26 Wirda2SICUbuntu nginx[46658]: nginx: [emerg] bind() to
<<[9E]> . 18 10:38:26 Wirda2SICUbuntu nginx[46658]: nginx: [emerg] bind() to
<<[9E]> . 18 10:38:26 Wirda2SICUbuntu nginx[46658]: nginx: [emerg] bind() to
<<[9E]> . 18 10:38:26 Wirda2SICUbuntu nginx[46658]: nginx: [emerg] bind() to
<<[9E]> . 18 10:38:27 Wirda2SICUbuntu nginx[46658]: nginx: [emerg] still coul
<<[9E]> . 18 10:38:27 Wirda2SICUbuntu systemd[1]: nginx.service: Control proc
<<[9E]> . 18 10:38:27 Wirda2SICUbuntu systemd[1]: nginx.service: Failed with
<<[9E]> . 18 10:38:27 Wirda2SICUbuntu systemd[1]: Failed to start A high perf
lines 1-18/18 (END)
[4]+  Stopped                  sudo systemctl status nginx
root@Wirda2SICUbuntu:~#
```

2. Instal paket curl menggunakan apt, kemudian gunakan dpkg untuk memastikan bahwa paket tersebut sudah benar-benar terpasang.

```
root@Wirda2SICUbuntu:~# sudo apt install curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-79-generic linux-hwe-6.8-headers-6.8.0-79
  linux-hwe-6.8-tools-6.8.0-79 linux-image-6.8.0-79-generic
  linux-modules-6.8.0-79-generic linux-modules-extra-6.8.0-79-generic
  linux-tools-6.8.0-79-generic
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  curl
0 upgraded, 1 newly installed, 0 to remove and 9 not upgraded.
Need to get 194 kB of archives.
After this operation, 455 kB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu jammy-updates/main amd64 curl amd64 7.
81.0-1ubuntu1.21 [194 kB]
Fetched 194 kB in 1s (181 kB/s)
Selecting previously unselected package curl.
(Reading database ... 243188 files and directories currently installed.)
Preparing to unpack .../curl_7.81.0-1ubuntu1.21_amd64.deb ...
Unpacking curl (7.81.0-1ubuntu1.21) ...
Setting up curl (7.81.0-1ubuntu1.21) ...
Processing triggers for man-db (2.10.2-1) ...
```

```

root@Wirda2SICUbuntu:~# dpkg -l | grep curl
ii  curl              7.81.0-1ubuntu1.21
    amd64          command line tool for transferring data with URL syntax
ii  libcurl3-gnutls:amd64 7.81.0-1ubuntu1.21
    amd64          easy-to-use client-side URL transfer library (GnuTLS flavour)
)
ii  libcurl4:amd64     7.81.0-1ubuntu1.21
    amd64          easy-to-use client-side URL transfer library (OpenSSL flavour)
r)
ii  php-curl           2:8.1+92ubuntu1
    all            CURL module for PHP [default]
ii  php8.1-curl        8.1.2-1ubuntu2.22
    amd64          CURL module for PHP

```

- Gunakan perintah dpkg untuk menampilkan versi dari paket mysql-server yang sudah terinstal di sistem.

```

root@Wirda2SICUbuntu:~# dpkg -s mysql-server | grep Version
Version: 8.0.43-0ubuntu0.22.04.2

```

- Tampilkan daftar semua paket yang mengandung kata php menggunakan dpkg, lalu arahkan hasilnya agar hanya menampilkan 10 baris pertama.

```

root@Wirda2SICUbuntu:~# dpkg -l | grep php | head -n 10
ii  libapache2-mod-php 2:8.1+92ubuntu1
    all            server-side, HTML-embedded scripting language (Apache 2 module) (default)
ii  libapache2-mod-php8.1 8.1.2-1ubuntu2.22
    amd64          server-side, HTML-embedded scripting language (Apache 2 module)
ii  php                 2:8.1+92ubuntu1
    all            server-side, HTML-embedded scripting language (default)
ii  php-common          2:92ubuntu1
    all            Common files for PHP packages
ii  php-curl            2:8.1+92ubuntu1
    all            CURL module for PHP [default]
ii  php-mbstring        2:8.1+92ubuntu1
    all            MBSTRING module for PHP [default]
ii  php-mysql           2:8.1+92ubuntu1
    all            MySQL module for PHP [default]
ii  php-xml             2:8.1+92ubuntu1
    all            DOM, SimpleXML, WDDX, XML, and XSL module for PHP [default]
ii  php8.1              8.1.2-1ubuntu2.22
    all            server-side, HTML-embedded scripting language (metapackage)
ii  php8.1-cli          8.1.2-1ubuntu2.22
    amd64          command-line interpreter for the PHP scripting language

```

- Buat arsip dari direktori /etc/ssh dengan nama ssh-backup.tar, kemudian kompres hasilnya menjadi ssh-backup.tar.gz.

```

root@Wirda2SICUbuntu:~# tar -cf ssh-backup.tar /etc/ssh
tar: Removing leading '/' from member names

```

- Buat arsip terkompresi dari direktori /var/log dengan nama log-backup.tar.gz, lalu tampilkan isi arsip tersebut tanpa mengekstrak.

```

root@Wirda2SICUbuntu:~# tar -czf log-backup.tar.gz /var/log
tar: Removing leading '/' from member names
root@Wirda2SICUbuntu:~# tar -tf log-backup.tar.gz
var/log/
var/log/dmesg.3.gz
var/log/mysql/
var/log/mysql/error.log
var/log/kern.log.1
var/log/mail.log.2.gz
var/log/dpkg.log.2.gz
var/log/auth.log.3.gz
var/log/dmesg.1.gz
var/log/auth.log.4.gz
var/log/syslog.2.gz
var/log/mail.log
var/log/speech-dispatcher/
var/log/boot.log.5
var/log/ubuntu-advantage.log.2.gz
var/log/boot.log.6
var/log/boot.log
var/log/mail.log.4.gz
var/log/vboxpostinstall.log
var/log/gdm3/
var/log/nginx/

```

7. Kompres file teks /var/log/syslog menggunakan gzip, lalu bandingkan ukuran file sebelum dan sesudah dikompresi.

```

root@Wirda2SICUbuntu:~# gzip /var/log/syslog
root@Wirda2SICUbuntu:~# ls -lh var/log/syslog*
ls: cannot access 'var/log/syslog*': No such file or directory
root@Wirda2SICUbuntu:~# ls -lh /var/log/syslog*
-rw-r----- 1 syslog adm 615K W.Ü. 18 09:29 /var/log/syslog.1
-rw-r----- 1 syslog adm 38K W.Ü. 11 07:15 /var/log/syslog.2.gz
-rw-r----- 1 syslog adm 40K W.Ü. 4 10:05 /var/log/syslog.3.gz
-rw-r----- 1 syslog adm 39K Ø.Ø. 21 09:53 /var/log/syslog.4.gz
-rw-r----- 1 syslog adm 26K W.Ü. 18 11:00 /var/log/syslog.gz

```